



## help binspwc

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### Title

**binspwc** — Data-Driven Nonparametric Pairwise Group Comparison using Binscatter.

### Syntax

```
binspwc depvar indvar [othercovs] [if] [in] [weight] , by(varname) [
2024
    estmethod(cmdname) deriv(v) at(position) nolink
    absorb(absvars) reghdfeopt(reghdfe_option)
    pwc(pwcopt) testtype(type) lp(metric)
    bins(p s) bynbins(bynbinsopt) binspos(position) binsmethod(method)
    nbinsrot(#) samebinsby randcut(#)
    pselect(numlist) sselect(numlist)
    nsims(#) simsgrid(#) simsseed(seed)
    dfcheck(n1 n2) masspoints(masspointsoption)
    vce(vcetype) asyvar(on/off) estmethodopt(cmd_option) usegtools(on/off) ]
```

where *depvar* is the dependent variable, *indvar* is the independent variable for binning, and *othercovs* are other covariates to be controlled for.

The degree of the piecewise polynomial *p*, the number of smoothness constraints *s*, and the derivative order *v* are integers satisfying  $0 \leq s, v \leq p$ , which can take different values in each case.

**fweights**, **awweights** and **pweights** are allowed; see [weight](#).

### Description

**binspwc** implements binscatter-based hypothesis testing procedures for pairwise group comparison of binscatter estimators, following the results in [Cattaneo, Crump, Farrell and Feng \(2024\)](#) and [Cattaneo, Crump, Farrell and Feng \(2026\)](#). If the binning scheme is not set by the user, the companion command [binsregselect](#) is used to implement binscatter in a data-driven (optimal) way and inference procedures are based on robust bias correction. Binned scatter plots based on different models can be constructed using the companion commands [binsreg](#), [binsqreg](#), [binslogit](#) and [binsprobit](#).

A detailed introduction to this command is given in [Cattaneo, Crump, Farrell and Feng \(2025\)](#). Companion R and Python packages with the same capabilities are available (see website below).

Companion commands: [binsreg](#) for binscatter least squares regression with robust inference procedures and plots, [binsqreg](#) for binscatter quantile regression with robust inference procedures and plots, [binslogit](#) for binscatter logit estimation with robust inference procedures and plots, [binsprobit](#) for binscatter probit estimation with robust inference procedures and plots, and [binsregselect](#) for data-driven (optimal) binning selection.

Related Stata, R and Python packages are available in the following website:

<https://nppackages.github.io/>

### Options

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Estimand

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**by**(*varname*) specifies the variable containing the group indicator to perform subgroup analysis; both numeric and string variables are supported. When **by**(*varname*) is specified, **binspwc** implements estimation for each subgroup separately and then conduct all pairwise comparison tests. By default, the binning structure is selected for each subgroup separately, but see the option **samebinsby** below for imposing a common binning structure across subgroups. This option is required.

**estmethod(cmdname)** specifies the binscatter model. The default is **estmethod(reg)**, which corresponds to the binscatter least squares regression. Other options are: **estmethod(qreg #)** for binscatter quantile regression where # is the quantile to be estimated, **estmethod(logit)** for binscatter logistic regression and **estmethod(probit)** for binscatter probit regression.

**deriv(v)** specifies the derivative order of the regression function for estimation, testing and plotting. The default is **deriv(0)**, which corresponds to the function itself.

**at(position)** specifies the values of *othercovs* at which the estimated function is evaluated for plotting. The default is **at(mean)**, which corresponds to the mean of *othercovs*. Other options are: **at(median)** for the median of *othercovs*, **at(0)** for zeros, and **at(filename)** for particular values of *othercovs* saved in another file.

Note: When **at(mean)** or **at(median)** is specified, all factor variables in *othercovs* (if specified) are excluded from the evaluation (set as zero).

**nolink** specifies that the function within the inverse link (logistic) function be reported instead of the conditional probability function. This option is used only if logit or probit model is specified in **estmethod()**.

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### Reghdfe

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**absorb(absvars)** specifies categorical variables (or interactions) representing the fixed effects to be absorbed. This is equivalent to including an indicator/dummy variable for each category of each *absvar*. When **absorb()** is specified, the community-contributed command **reghdfe** instead of the command **regress** is used.

**reghdfeopt(reghdfe\_option)** options to be passed on to the command **reghdfe**. Important: **absorb()** and **vce()** should not be specified within this option.

For more information about the community-contributed command **reghdfe**, please see <http://scorreia.com/software/reghdfe/>.

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### Pairwise Group Comparison Testing

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**pwc(pwcopt)** sets the degree of polynomial and the number of smoothness constraints for pairwise group comparison. If **pwc(p s)** is specified, a piecewise polynomial of degree *p* with *s* smoothness constraints is used. If **pwc(T)** or **pwc()** is specified, **pwc(1 1)** is used unless the degree *p* or smoothness *s* selection is requested via the option **pselect()** or **sselect()** (see more details in the explanation of **pselect()** and **sselect()**). The default is **pwc()**.

**testtype(type)** specifies the type of pairwise comparison test. The default is **testtype(2)**, which corresponds to a two-sided test of the form  $H_0: \mu_1(x) = \mu_2(x)$ . Other options are: **testtype(1)** for the one-sided test of the form  $H_0: \mu_1(x) \leq \mu_2(x)$  and **testtype(r)** for the one-sided test of the form  $H_0: \mu_1(x) \geq \mu_2(x)$ .

**lp(metric)** specifies an  $L_p$  metric used to test for the difference between two groups. The default is **lp(inf)**, which corresponds to the sup-norm. Other options are **lp(q)** for a positive number *q* no less than 1. Note that **lp(inf)** ("sup norm") has to be used for one-sided tests.

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### Binning/Degree/Smoothness Selection

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**bins(p s)** sets a piecewise polynomial of degree *p* with *s* smoothness constraints for data-driven (IMSE-optimal) selection of the partitioning/binning scheme. The default is **bins(0 0)**, which corresponds to the piecewise constant.

**bynbins**(*bynbinsopt*) sets the number of bins for partitioning/binning of *indvar*. If **bynbins**(*numlist*) is specified, the number in the *numlist* is applied to the *binscatter* estimation for each group. The ordering of the group follows the result of *tabulate*. If a single number of bins is specified, it applies to the estimation for all groups. If **bynbins**(**T**) or **bynbins**() (default) is specified, the number of bins is selected via the companion command *binsregselect* in a data-driven, optimal way whenever possible.

Note: If a *numlist* with more than one number is supplied within **bynbins**(), it is understood as the number of bins applied to *binscatter* estimation for each subgroup rather than the range for selecting the number of bins.

**binspos**(*position*) specifies the position of binning knots. The default is **binspos**(**qs**), which corresponds to quantile-spaced binning (canonical *binscatter*). Other options are: **es** for evenly-spaced binning, or a *numlist* for manual specification of the positions of inner knots (which must be within the range of *indvar*).

**binsmethod**(*method*) specifies the method for data-driven selection of the number of bins via the companion command *binsregselect*. The default is **binsmethod**(**dpi**), which corresponds to the IMSE-optimal direct plug-in rule. The other option is: **rot** for rule of thumb implementation.

**nbinsrot**(*#*) specifies an initial number of bins value used to construct the DPI number of bins selector. If not specified, the data-driven ROT selector is used instead.

**samebinsby** forces a common partitioning/binning structure across all subgroups specified by the option **by**(). The knots positions are selected according to the option **binspos**() and using the full sample. If **nbins**() is not specified, then the number of bins is selected via the companion command *binsregselect* and using the full sample.

**randcut**(*#*) specifies the upper bound on a uniformly distributed variable used to draw a subsample for bins/degree/smoothness selection. Observations for which **runiform**()<=*#* are used. *#* must be between 0 and 1. By default, max(5000, 0.01*n*) observations are used if the samples size *n*>5000.

**pselect**(*numlist*) specifies a list of numbers within which the degree of polynomial *p* for point estimation is selected. If the selected optimal degree is *p*, then piecewise polynomials of degree *p*+1 are used to conduct pairwise group comparison.

**sselect**(*numlist*) specifies a list of numbers within which the number of smoothness constraints *s* for point estimation is selected. If the selected optimal smoothness is *s*, then piecewise polynomials with *s*+1 smoothness constraints are used to conduct pairwise group comparison. If not specified, for each value *p* supplied in the option **pselect**(), only the piecewise polynomial with the maximum smoothness is considered, i.e., *s*=*p*.

Note: To implement the degree or smoothness selection, in addition to **pselect**() or **sselect**(), **bynbins**(*numlist*) must be specified.

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### Simulation

**nsims**(*#*) specifies the number of random draws for hypothesis testing. The default is **nsims**(**500**), which corresponds to 500 draws from a standard Gaussian random vector of size [(*p*+1)\**J* - (*J*-1)\**s*]. Setting at least **nsims**(**2000**) is recommended to obtain the final results.

**simsgrid**(*#*) specifies the number of evaluation points of an evenly-spaced grid within each bin used for evaluation of the supremum (infimum or *Lp* metric) operation needed to construct confidence bands and hypothesis testing procedures. The default is **simsgrid**(**20**), which corresponds to 20 evenly-spaced evaluation points within each bin for approximating the supremum (infimum or *Lp* metric) operator. Setting at least **simsgrid**(**50**) is recommended to obtain the final results.

**simsseed**(*#*) sets the seed for simulations.

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### Mass Points and Degrees of Freedom

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**dfcheck**(*n1 n2*) sets cutoff values for minimum effective sample size checks, which take into account the number of unique values of *indvar* (i.e., adjusting for the number of mass points), number of clusters, and degrees of freedom of the different statistical models considered. The default is **dfcheck(20 30)**. See Cattaneo, Crump, Farrell and Feng (2025) for more details.

**masspoints**(*masspointsoption*) specifies how mass points in *indvar* are handled. By default, all mass point and degrees of freedom checks are implemented.

Available options:

**masspoints**(*noadjust*) omits mass point checks and the corresponding effective sample size adjustments.

**masspoints**(*nolocalcheck*) omits within-bin mass point and degrees of freedom checks.

**masspoints**(*off*) sets **masspoints**(*noadjust*) and **masspoints**(*nolocalcheck*) simultaneously.

**masspoints**(*veryfew*) forces the command to proceed as if *indvar* has only a few number of mass points (i.e., distinct values). In other words, forces the command to proceed as if the mass point and degrees of freedom checks were failed.

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### Other Options

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**vce**(*vcetype*) specifies the *vcetype* for variance estimation used by the commands **regress**, **logit**, **probit**, **greg** or **reghdfe**. The default is **vce(robust)**.

**asyvar**(*on/off*) specifies the method used to compute standard errors. If **asyvar**(*on*) is specified, the standard error of the nonparametric component is used and the uncertainty related to other control variables *othercovs* is omitted. Default is **asyvar(off)**, that is, the uncertainty related to *othercovs* is taken into account.

**estmethodopt**(*cmd\_option*) options to be passed on to the estimation command specified in **estmethod()**. For example, options that control for the optimization process can be added here.

**usegtools**(*on/off*) forces the use of several commands in the community-distributed Stata package **gtools** to speed the computation up, if *on* is specified. Default is **usegtools(off)**.

For more information about the package **gtools**, please see <https://gtools.readthedocs.io/en/latest/index.html>.

### Examples

```
Setup
. sysuse auto

Generate two groups
. gen group=price>5000

Test for the difference between two groups
. binspwc mpg weight foreign, by(group)
```

### Stored results

#### Scalars

|             |                                            |
|-------------|--------------------------------------------|
| <b>e(N)</b> | number of observations                     |
| <b>e(p)</b> | degree of polynomial for bin selection     |
| <b>e(s)</b> | smoothness of polynomial for bin selection |

#### Macros

|                   |                                     |
|-------------------|-------------------------------------|
| <b>e(byvalue)</b> | name of groups found in <b>by()</b> |
|-------------------|-------------------------------------|

#### Matrices

|                     |                                            |
|---------------------|--------------------------------------------|
| <b>e(N_by)</b>      | number of observations for each group      |
| <b>e(Ndist_by)</b>  | number of distinct values for each group   |
| <b>e(Nclust_by)</b> | number of clusters for each group          |
| <b>e(nbins_by)</b>  | number of bins for each group              |
| <b>e(pwc_plist)</b> | degree of polynomial for testing, by group |

|                               |                                                |
|-------------------------------|------------------------------------------------|
| <code>e(pwc_slist)</code>     | smoothness of polynomial for testing, by group |
| <code>e(stat)</code>          | test statistics for all pairwise comparisons   |
| <code>e(pval)</code>          | p values for all pairwise comparisons          |
| <code>e(imse_var_rot)</code>  | variance constant in IMSE, ROT selection       |
| <code>e(imse_bsqr_rot)</code> | bias constant in IMSE, ROT selection           |
| <code>e(imse_var_dpi)</code>  | variance constant in IMSE, DPI selection       |
| <code>e(imse_bsqr_dpi)</code> | bias constant in IMSE, DPI selection           |

## **References**

Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng. 2024. [On Binscatter](#). *American Economic Review* 114(5): 1488-1514.

Supplemental Appendix for On Binscatter. [Supplemental Appendix](#).

Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng. 2026. [Nonlinear Binscatter Methods](#). *Review of Economics and Statistics*, revise and resubmit.

Supplemental Appendix for Nonlinear Binscatter Methods. [Supplemental Appendix](#).

Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng. 2025. [Binscatter Regressions](#). *Stata Journal* 25(1): 3-50.

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